Sign recovery and centered norms for explicit ZFC consistency equations

Research contribution prepared for the ProveIt project

23 October 2026

Abstract

We give explicit improvements to several Diophantine equations calibrated to the syntactic consistency of ZFC. The main short equation has thirteen strictly positive integer unknowns, exact total degree 114, and an 83-operation division-free evaluator. Centering an algebraic norm at a known large part of a Pell root replaces two root witnesses by one centered witness with lower degree. A separate common-sign argument gives a quartic in forty-two unrestricted integers: two independent positive representations can be replaced by one positive representation of their product, because a larger existing square scale lets the Pell indices recover the missing sign. A proved exchange of scale guards followed by exact eliminations gives a sextic in thirty-five integers and an octic in thirty integers. A companion sparse packing optimization gives nine positive unknowns, degree 1 834 516, and 453 operations.

Each equation comes with a literal polynomial or arithmetic circuit, fixed coefficient payloads, a two-way ordinary mathematical reduction, and finite verification programs. The comparisons use the pinned ProveIt revision stated below. The native computation compiler and K215 calibration are inherited. We do not claim a globally simplest equation, an independent literature priority result, or a new end-to-end formal proof. These are concrete local improvements, not an established major breakthrough in Diophantine representation.

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1 Results, comparisons, and the exact logical claim

The most readable new short equation is $P_{13} = 0$ in Section 3. It uses thirteen positive inputs and four *fixed* integer coefficients. Its degree is 114 and its evaluator uses 83 binary additions, subtractions, and multiplications, including all scalar multiplications and squarings. The lowest-degree new equation is the quartic $Q_{42} = 0$ in Section 4; all forty-two of its unknowns range freely over $\mathbb{Z}$.

The comparison source is the repository `VladimirReshetnikov/ProveIt`, directory `Computability/DiophantineConsistency/con-zfc-diophantine-forms`, at the commit

`beb0192cb8dc36f78c369c080f2adfbc1887a486`.

This includes the six wave-eight packages numbered 54 through 59. In particular, the older collection index alone is not the comparison frontier. The relevant late package READMEs and exact predecessor objects are retained or pinned in the accompanying archive [4, 5, 6].

Domain	Construction	Unknowns	Degree	Binary operations
$\mathbb{Z}_{>0}$	Source 59 short adjunct	13	126	91
$\mathbb{Z}_{>0}$	Source 57 short form	13	132	83
$\mathbb{Z}_{>0}$	**Centered norm, this work**	**13**	**114**	**83**
$\mathbb{Z}$	Source 57 quartic	45	4	Not optimized
$\mathbb{Z}$	**Common-sign quartic**	**42**	**4**	Not optimized
$\mathbb{Z}$	Archived sextic, sources 27/29	42	6	Not optimized
$\mathbb{Z}$	**New sextic companion**	**35**	**6**	Not optimized
$\mathbb{Z}$	Archived octic, sources 27/29	36	8	Not optimized
$\mathbb{Z}$	**New octic companion**	**30**	**8**	Not optimized
$\mathbb{Z}_{>0}$	Source 59 sparse form	9	1 910 404	473
$\mathbb{Z}_{>0}$	**Repacked companion**	**9**	**1 834 516**	**453**

Table 1: Exact comparisons within each domain and degree or variable-count regime. The old sextic and octic are historical low-degree baselines, not the fewest-variable unrestricted-integer equations in the collection.

There are additional useful tradeoffs in the repository. For example, source 58 has a fifteen-positive-variable equation of degree 24, and source 57 has twelve-positive-variable equations of degree 242 and 258 [4, 5]. This paper does not improve those points. The different domains in Table 1 must not be conflated: thirteen positive variables and thirteen unrestricted integer variables are different representation problems. No nine-unrestricted-variable claim is made here.

1.1 The theorem being proved

Fix the repository's effective presentation of ZFC and write $\mathrm{Ref}_{\mathrm{ZFC}}(p)$ for its primitive recursive proof-checking predicate for a refutation. Define the syntactic consistency statement by

$$\mathrm{Con(ZFC)} \;\equiv\; \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

For every polynomial P in Table 1 introduced by this work, and its stated domain D, the claimed mathematical theorem is

$$\mathrm{ZFC} \;\vdash\; (\exists \boldsymbol{x} \in D^n\ P(\boldsymbol{x}) = 0) \longleftrightarrow (\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p))\,. \tag{1.1}$$

Negation gives precisely the requested equivalence

$$\mathrm{ZFC} \;\vdash\; (\neg\exists \boldsymbol{x} \in D^n\ P(\boldsymbol{x}) = 0) \longleftrightarrow \mathrm{Con(ZFC)}. \tag{1.2}$$

Neither implication assumes that ZFC is consistent. A proof of (1.2) is also not a proof in ZFC that P has no zero. The latter would prove the consistency statement itself.

The new steps are arithmetic: integer polynomial identities, Pell recurrences and classification, divisibility, elementary bounds, and finite coefficient constructions. They are ordinary mathematics formalizable in ZFC. Their composition uses the expressly identified native compiler and literal K215-to-refutation calibration. Thus the new constructions remain based on the inherited K215 endpoint. The quartic, sextic, octic, and thirteen-variable equation do not need a Jones or Sun variable-reduction theorem. The nine-variable companion retains its Jones ratio interface.

1.2 What "simpler" measures here

We report the domain, number of quantified unknowns, exact total degree, arithmetic operation count when optimized, and fixed coefficient size separately. An arithmetic register is an abbreviation, not an extra existential unknown. Every exponent in an emitted polynomial circuit is a fixed numeral, and every power is expanded to charged multiplication gates. Constants are leaves; reusing a computed register is free. This counts arithmetic operations, not bit operations on the very large coefficients or witnesses.

The thirteen-variable result strictly improves both displayed predecessor points in degree without increasing either operation count. The common-sign construction reduces the number of integer unknowns at fixed quartic degree. The nine-variable result improves its predecessor by 75 888 degrees and 20 operations. None of these finite comparisons establishes a global minimum. MRDP establishes existence of appropriate Diophantine representations [1, 2]; it does not certify an optimal explicit instance. Also, current unrestricted-integer variable-count results must not be summarized by the older eleven-variable record: Sun's September 2026 preprint states a ten-variable undecidability theorem [3]. That universal theorem is background, not an additional reduction silently applied to the equations here.

2 Fixed coefficients and inherited dependencies

A display containing four unexplained coefficient symbols would specify only a family. Here those symbols are bound to complete finite byte strings, together with deterministic reconstruction programs and their input data. The fixed integers are unsigned, minimally encoded in little-endian order. The compressed payloads carry no mathematical ambiguity: their decompressed sizes and cryptographic digests are checked. The native transition table and tile definitions used to reconstruct them are included.

Whole native tuple	β bits	K_0 bits	M_C bits	M_F bits
Compact 44: P_{13} and nine	590069	592884	590069	590070
Source 34: Q_{42} and companions	1064921	1065043	1064921	1064922

Table 2: Bit lengths of the fixed, coherent native tuples. The full manifests and all four numeral payloads accompany each construction.

For the compact tuple, $\beta = 2^{590069} - 1$ and the four raw integer encodings occupy 295 388 bytes. For the quartic tuple, $\beta = 2^{1064921} - 1$ and they occupy 532 479 bytes. The fourth port M_F is the *already shifted* source mask in both presentations. The exponent of β alone is not an instance specification, and individual ports from different tuples must not be mixed. The nine-variable final evaluator has twelve fixed leaves derived from its native tuple; their encodings total 2 217 161 raw bytes and 47 364 gzip bytes. Those final leaves are supplied too.

The dependency chain has three mathematical layers. At the bottom, a literal K215 halting

computation is equivalent to a refutation in the fixed ZFC calculus. The native window compiler and its cofinal padding turn that halting computation into positive solutions of the bound arithmetic source system and decode every source solution back to such a computation. Finally, the new polynomial transformations in this paper preserve existence of solutions in both directions. The archive retains the ordinary proofs used at each interface and records exact repository paths, commit pins, Git blob identities and SHA256 digests. The detailed reading map is `quartic42/dependencies/README.md`.

The included formal endpoint is

`KEndpoint215.blank_halts_iff_zfc_refutation`.

It identifies blank halting with the repository's `BProv ZFCax_s [] Form.fBot`. Its complete transitive Lean/Rocq development is not rebuilt in this archive. More importantly, an inherited machine theorem is not a new kernel certificate for the literal equations introduced here. We provide ordinary mathematical reductions and exact finite data checks; we do not provide an exported object-language ZFC derivation or a fresh end-to-end proof-assistant build. This boundary is part of the result, not an unstated hypothesis that ZFC is consistent.

3 A root-centered equation in thirteen positive integers

The construction in this section has thirteen strictly positive integer unknowns, exact total degree 114, and 83 live binary arithmetic operations: 46 multiplications, 19 additions, and 18 subtractions. Squaring and multiplication by a fixed scalar are charged. Intermediate registers are polynomial abbreviations and do not add unknowns.

The new step merges the first Pell root and the auxiliary Pell ordinate using a centered two-root norm. Keeping the ordinary strong coordinate allows direct recovery of both deleted roots. This gives a degree improvement over the pinned thirteen-positive constructions of degree 126 in 91 operations and degree 132 in 83 operations. These are comparisons with the named repository constructions, without a claim of global minimality or literature priority.

3.1 The fixed equation

Use the entire compact44 native tuple, with $\beta = 2^{590069} - 1$. The four payloads and their exact hashes are fixed by `COEFFICIENT_BINDING.json`; the aliases are $\beta \mapsto \beta$, $K_0 \mapsto K$, $M_C \mapsto M_C$, and $M_F \mapsto M_{F,\mathrm{source}}$. The last coefficient is already shifted: $M_{F,\mathrm{source}} = M_{F,\mathrm{native}} + 2^{590069} - 1$. The whole tuple has 295388 raw bytes. Its included proof supplies the same native blank-machine compiler interface, with the ordinary strong witness obtained from a canonical normalized one by $i_{\mathrm{ordinary}} = \Delta i_{\mathrm{normalized}}$.

Supply precisely

$$J, F, \alpha, z, f, i, T, s, w, \eta, \zeta, \gamma, t \quad \in \mathbb{Z}_{>0}.$$

Define the following polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & & \\
X &= wq, & Y &= sq^3, & E &= XY, & k &= \eta + \zeta, \\
c &= kY + \eta, & a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H, \\
\ell &= EkY, & A_1 &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2, & b_0 &= X + ac.
\end{aligned}$$

The packing and transport expressions are

$$\begin{aligned}
R &= \big((q-1)(q-F) + \alpha\big)(q^2 - 1) + (M_C + qM_F)J, \\
N_t &= (K_0 + w)C + q - F - z(q-1).
\end{aligned}$$

For the absorbed main root put

$$U = 4q^4, \qquad r = k - R - 1,$$
$$D_* = Ub_0 + H(U\gamma - r), \qquad N_{\mathrm{main}} = D_*^2 - U^2 G.$$

The strong and auxiliary quantities are

$$I = ic^2, \quad Q = \Delta(f^2-1), \quad d = Q-1, \quad N_s = I^2 - d,$$
$$V = c(Tf-1) - Rf^2, \quad E_{\mathrm{aux}} = QV^2 - 1,$$
$$x = t + \ell, \quad \delta = x^2 - A_1, \quad N = (d\delta - E_{\mathrm{aux}})^2 - 4dA_1E_{\mathrm{aux}}.$$

The proposed equation is

$$\boxed{P_{13} = N_t(1+N_{\mathrm{main}})N_s(1+N) - 1 = 0.} \tag{3.1}$$

Every displayed power has a fixed nonnegative integer exponent. The included division-free JSON circuit expands every abbreviation. The coordinate z here is the old positive *product* transport coordinate. It equals $z_{\mathrm{shift}}+1$ for the coordinate used in a shifted equation $(K_0+w)C = F + z_{\mathrm{shift}}(q-1)$.

With the fixed tuple, ordinary mathematics formalizable in ZFC proves

$$\exists \boldsymbol{v} \in \mathbb{Z}_{>0}^{13}\ P_{13}(\boldsymbol{v}) = 0 \quad \Longleftrightarrow \quad \text{there is a finite syntactic ZFC refutation.}$$

Negating this equivalence gives the required equivalence between the absence of positive zeros and Con(ZFC). The native compiler and the literal K215 calibration are inherited theorems; no new end-to-end proof-assistant certificate is claimed.

3.2 The centered two-root identity

For arbitrary indeterminates d, x, A, e there are exact polynomial identities

$$\begin{aligned}(d(x^2-A)-e)^2 - 4dAe &= \big(d(x^2+A)-e\big)^2 - 4d^2x^2A \\ &= d^2(x^2-A)^2 - 2d(x^2+A)e + e^2.\end{aligned} \tag{3.2}$$

The first expression saves one multiplication in the evaluator. If d, x, A are positive integers and it vanishes, the middle expression gives the rational number

$$\tau_0 = \frac{d(x^2+A)-e}{2dx}, \qquad \tau_0^2 = A.$$

A rational square root of an integer is an integer. Therefore $\tau_0 \in \mathbb{Z}$ and

$$d(x-\tau_0)^2 = e. \tag{3.3}$$

This direct argument allows coincident square classes. It requires no unproved independence of formal radical signs and no auxiliary rational Pell saturation lemma.

The particular shift $x = t + \ell$ gives

$$\delta = t^2 + \ell(2t-k) - 1. \tag{3.4}$$

This cancellation lowers its degree from the apparent 22 to 12. The converse will prove positivity of the new witness $t = \tau - \ell + y$; positivity is not assumed to be preserved by an arbitrary affine change of an old tuple.

3.3 The unit guard and the two restored roots

At a positive zero of (3.1), each of the four integer factors is 1 or -1. Since $4 \mid U$, the residue of N_{main} modulo 4 is the square D_*^2. The residue of N modulo 4 is also a square. Neither residual can be -2. Thus $N_{\mathrm{main}} = N = 0$ and their two factors are $+1$.

For every integer a, $\Delta = a^2 + 4a + 3$ is 0 or 3 modulo 4. Hence $Q = \Delta(f^2 - 1)$ is 0 or 1 modulo 4. It follows that $N_s = 1 + I^2 - Q$ is 0, 1, or 2 modulo 4 and cannot be -1. Consequently

$$Q = I^2, \qquad N_t = 1, \qquad N_{\mathrm{main}} = N = 0. \tag{3.5}$$

Here $c > 1$ and $I = ic^2 > 1$, so $Q > 1$ and $d > 0$. In particular the apparently troublesome case $f = 1$ is excluded by the equations themselves. Also $x = t + \ell > 0$ and $A_1 > 0$.

If $V = 0$, then $E_{\mathrm{aux}} = -1$, and

$$N = (d\delta + 1)^2 + 4dA_1 > 0,$$

a contradiction. Therefore $V \neq 0$ and $E_{\mathrm{aux}} > 0$. Applying (3.2) and (3.3) gives integers τ_0 and $y_0 = x - \tau_0$ with

$$\tau_0^2 = A_1, \qquad dy_0^2 = E_{\mathrm{aux}}.$$

Thus the positive integers $\tau = |\tau_0|$ and $y = |y_0|$ satisfy

$$\tau^2 - \ell(\ell + k) = 1, \qquad QV^2 - (Q - 1)y^2 = 1, \qquad (ic^2)^2 = \Delta(f^2 - 1). \tag{3.6}$$

The identity $\ell(\ell + k) = (XY^2)(XY^2 + 1)k^2$ identifies the first original Pell norm. Together with (3.5), these are precisely the ordinary-strong fourteen-coordinate absorber equations, with all retained coordinates still positive. Neither an index quotient nor its sign has yet been assumed.

3.4 Early index recovery and restoration of the compiler

The original absorber proof is supplied verbatim as `upstream/FOURTEEN_ABSORPTION_PROOF.md`; its full source identity is recorded in the manifest. We restate its proof order because the early index bound is necessary for the ordinary strong-rank lemma.

The equality $N_t = 1$ implies $C > 0$: if $C \leq 0$, positivity of $K_0 + w$ gives

$$N_t \leq q - F - z(q - 1) = 1 - F - (z - 1)(q - 1) \leq 0.$$

The authentic mask bounds then give

$$q \geq 16, \quad 3q < R < q^4 - 2, \quad E \geq q^4 > R + 2, \quad kY < c < k(Y + 1), \quad a + 2 < 2E. \tag{3.7}$$

From $N_{\mathrm{main}} = 0$, $g = D_*/U$ is a rational square root of the integer G, hence an integer. It is positive. Indeed, $H < 5a$, $b_0 > akY$, $r < k$, and $\gamma > 0$ imply

$$D_* > Ub_0 - Hk > ak(UY - 5) > 0.$$

Thus $g^2 - \Delta c^2 = 1$. Reduction of $D_* = Ug$ modulo U gives

$$U \mid H(k - R - 1). \tag{3.8}$$

Put $A_0 = a + 2$ and $P_0 = 2XY^2 + 1$. The usual positive Pell sequences give indices $p, n \geq 1$ such that

$$g = \chi_{A_0}(p), \quad c = \psi_{A_0}(p), \qquad \tau = \chi_{P_0}(n), \quad k = 2\psi_{P_0}(n).$$

Since $P_0 > A_0$ and $c > k$, monotonicity implies $p > n$. The first recurrence gives $k \equiv 2n \pmod{4XY^2}$, and $U \mid 4XY^2 = 4ws^2q^7$. Moreover H is odd and $H \equiv 3 \pmod q$; if $3 \mid q$, then $H \equiv 3 \pmod{27}$. Hence $\gcd(U, H) \le 3$. Defining $\mu = U/\gcd(U, H)$, we obtain

$$\mu \ge \frac{4q^4}{3} > R + 1, \qquad 2n \equiv R + 1 \pmod{\mu}, \qquad n \ge \frac{R+1}{2} \ge 25.$$

The lower bound precedes recovery of $p = R$. It gives $p \ge 26$, and Pell growth therefore proves

$$c > A_0\Delta^2, \qquad c > 2p, \qquad c > 2R. \tag{3.9}$$

For example, $c \ge (2A_0 - 1)^{p-1} \ge A_0^{25} > A_0(A_0^2 - 1)^2$; also $c \ge \psi_{A_0}(2) = 2A_0 > 2R$.

The ordinary strong-rank lemma in the supplied `ORDINARY_BLANK_PROOF.md`, Section 4, now applies. It yields

$$\begin{gathered} f = \chi_{A_0}(m), \quad p \mid m, \quad c \mid m, \quad m \ge c > 2p, \\ I = \Delta\psi_{A_0}(m), \qquad c^2 \mid (f^2 - 1). \end{gathered} \tag{3.10}$$

This lemma treats the possible nonfundamentality of A_0 in the underlying Pell group. It is used with the verified strict hypothesis (3.9); it is not replaced by informal cancellation of Δ in the strong equation.

The auxiliary equation in (3.6), the congruence $V \equiv -c \pmod f$, and the odd abscissa quotient identity give an odd auxiliary index j_0 with $j_0 \equiv \pm p \pmod m$. On the other hand $Q \equiv 0 \pmod c$, $f^2 \equiv 1 \pmod c$, and $V \equiv -R \pmod c$ give $j_0 \equiv \pm R \pmod c$. The signed index theorem from the ordinary source, Section 5, applies because $0 < 2p < m$. Since $c \mid m$ and $0 < p, R < c/2$, it follows that $p = R$. This step permits either sign of V.

Now $n < p = R$, so $|2n - R - 1| < R < \mu$, and the same congruence forces $2n = R + 1$. Consequently $r = k - 2n$ is divisible by $4XY^2$, hence by both E and U. Set

$$h = \frac{r}{E}, \qquad \gamma_{\mathrm{old}} = \gamma - \frac{r}{U}.$$

The first is a positive integer because $\psi_{P_0}(n) > n$ for $n \ge 25$. The second equals $(g - b_0)/H$. Since $n \ge 2$, $k \ge 4P_0 > X$ and $c > X$; furthermore $g > (a + 1)c$, so $g - b_0 > c - X > 0$. Thus every original ordinary compiler coordinate has been restored with its required positive sign. The inherited compiler soundness gives the blank K215 halt and hence a finite ZFC refutation.

3.5 Constructive completeness

Conversely, start with a canonical positive ordinary compiler witness of a blank K215 halt. Canonical normalized witnesses supply these by $i_{\mathrm{ordinary}} = \Delta i_{\mathrm{normalized}}$. Its index theorem gives $p = R = 2n - 1$, so $r = k - 2n$ is a positive multiple of $4XY^2$ and therefore of U. Define

$$\gamma = \gamma_{\mathrm{old}} + \frac{r}{U}, \qquad t = \tau - \ell + y.$$

Both are positive integers. In particular $\tau^2 = 1 + \ell(\ell + k) > \ell^2$ implies $\tau > \ell$. Retain all other supplied coordinates. Then $x = \tau + y$, $A_1 = \tau^2$, and $E_{\mathrm{aux}} = dy^2$, so (3.2) gives $N = 0$. Also $D_* = Ug$, giving $N_{\mathrm{main}} = 0$, while $N_t = N_s = 1$ are unchanged. Therefore (3.1) holds. This proves both existential directions, with no additional cofinality condition on an auxiliary index.

3.6 Exact degree and finite verification

All witnesses have degree one and the four ports degree zero. Put

$$q_0 = \beta J, \qquad E_0 = wsq_0^4, \qquad c_0 = (\eta + \zeta)sq_0^3, \qquad \ell_0 = E_0c_0.$$

Their degrees are $1, 6, 5, 11$, respectively. In the inner norm,

$$\deg d = 14, \quad \deg V = 7, \quad \deg E_{\mathrm{aux}} = 28, \quad \deg A_1 = 22, \quad \deg \delta = 12.$$

The three terms of the final expression in (3.2) have degrees $52, 64, 56$. Only the middle one reaches degree 64, with homogeneous part

$$N_{64} = -4E_0^6 c_0^4 T^2 f^6. \tag{3.11}$$

In the efficient evaluator the squared core has degree at most 56; its correction $-4dA_1E_{\mathrm{aux}}$ has exact degree 64.

The main residual has the exact identity

$$\begin{aligned} N_{\mathrm{main}} = U^2\big[2ac(X + H\gamma) + (X + H\gamma)^2 - Hc^2 - 1\big] \\ - 2UHr(X + ac + H\gamma) + H^2r^2. \end{aligned}$$

Its three displayed terms have degrees 26, at most 25, and at most 20, so

$$(N_{\mathrm{main}})_{26} = 128q_0^8 E_0^2 c_0 \gamma.$$

The other factors have leaders

$$(N_s)_{22} = i^2 c_0^4, \qquad (N_t)_2 = w(q_0 - F - \alpha) - zq_0.$$

Thus $\deg P_{13} \le 2 + 26 + 22 + 64 = 114$. On the full diagonal, where every witness is replaced by one indeterminate, the four leading coefficients are

$$-2, \qquad 256\beta^{19}, \qquad 16\beta^{12}, \qquad -64\beta^{36}.$$

Their product is $2^{19}\beta^{67} \neq 0$ on the fixed coefficient slice. This proves exact degree 114 without treating the ports as additional variables or using a numerical experiment as an upper-degree proof.

The primary circuit uses 83 binary gates, all live. The independent standard-library checker imports no builder. It authenticates the whole tuple and all source pins; independently verifies 41 named formulas against the literal circuit; proves the norm, centering, and main cancellation identities; checks the universal weighted degree bound; and evaluates six complete polynomial rays over three finite fields using the actual fixed tuple. Normal and optimized Python runs agree. These finite checks verify transcription, fixed data, and degree. They do not replace the unbounded mathematical compiler and calibration proofs.

The original source50 absorber, ordinary strong-rank proof, compact native adapter, and source59 comparison are preserved in the included upstream directory. Their exact source paths, commits, Git blob identities and SHA256 digests are recorded in `SOURCE_PINS.json`. The complete formal-assistant dependency tree of the inherited K215 endpoint is not claimed to have been rebuilt in this work.

4 A quartic equation in forty-two unrestricted integers

The new equation in this section has *forty-two unrestricted integer unknowns*, exact total degree four, and twenty-seven quadratic residuals. It improves the supplied forty-five-integer quartic without increasing its degree. The four fixed coefficient ports are the complete, unchanged tuple of that predecessor. In particular, the high-mask port is already shifted. The construction uses the direct Pell/compiler route; no Jones or Sun variable-reduction theorem is needed for this result.

Theorem. With the fixed coefficients specified in the accompanying binding, the polynomial Q_{42} defined below satisfies

$$\mathrm{ZFC} \;\vdash\; \big[\neg\exists \boldsymbol{x} \in \mathbb{Z}^{42}\; Q_{42}(\boldsymbol{x}) = 0\big] \;\longleftrightarrow\; \mathrm{Con}(\mathrm{ZFC}).$$

The argument is an ordinary arithmetic proof, composed with the expressly retained native compiler and machine-calibration theorems. The new literal polynomial has not received an end-to-end Lean or Rocq certificate. An exported derivation in the repository's object-language ZFC calculus is also distinct from the mathematical proof given here.

4.1 The idea: prove the sign after recovering the rank

Two positive coordinates in the predecessor, η and ζ, are each represented by a ternary positive form. Those two representations use six extra integer unknowns. We replace them by one representation of $\eta\zeta > 0$, which uses three. This initially proves only that the two coordinates have the same sign.

An existing square scale supplies the missing information. Replacing $w = u^2$ by $w = 4u^2$ makes a modulus e more than four times the decoded index R. The Pell equations first recover a rank $p = R$ without deciding the common sign. A negative common sign would then require a positive integer smaller than e to be divisible by e. Thus both coordinates are positive after all. Canonical accepting witnesses have w equal to a sufficiently large dyadic square, so the factor four does not obstruct completeness.

This ordering is essential. Invoking the predecessor's whole soundness theorem before recovering the missing positive guards would be circular. The local rank argument below is what permits those guards to be removed.

4.2 Every variable and every residual

Write the four fixed positive integer ports as β, K_0, M_C, M_F, where $\beta = 2^{1064921} - 1$ and M_F denotes the *already shifted* source mask. The complete coefficient binding, not that exponent alone, specifies the instance. Let the thirty-three arithmetic unknowns be

$$J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, y, \gamma, o, j,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_*, \Delta, u, \rho.$$

Add nine guard unknowns $A_1, A_2, A_3, B_1, B_2, B_3, T_1, T_2, T_3$. All forty-two range over $\mathbb{Z}$. In particular, C_2, F_2, B_* and Δ are independent supplied coordinates; their defining equations are among the residuals. Define only the following abbreviations:

$$q = \beta J + 1, \qquad C = q - F - \alpha, \qquad k = \eta + \zeta, \qquad \mathrm{Pos}(x, y, t) = 1 + x^2 + y^2 + t(t+1).$$

These abbreviations add no quantified unknowns. The residuals are

$$\begin{aligned}
R_1 &= L - q^2 + 1, & R_2 &= N - q(L+1),\\
R_3 &= X - wq, & R_4 &= a - E - (2v+1)N,\\
R_5 &= E(X+1) - Xa, & R_6 &= c - \eta - (a-E)k,\\
R_7 &= P - k - E(c-\eta), & R_8 &= \tau^2 - P(P-k) - 1,\\
R_9 &= D - X - ac - \gamma(4a+3). & &
\end{aligned} \tag{4.1}$$

The next block is

$$\begin{aligned}
R_{10} &= \Delta - a^2 - 4a - 3,\\
R_{11} &= C_2 - c^2,\\
R_{12} &= D^2 - \Delta C_2 - 1,\\
R_{13} &= k - 1 - hE - \big[(q-F)(N-q) - CL + (M_C + qM_F)J\big],\\
R_{14} &= (K_0 + w)C - F - z(q-1),\\
R_{15} &= F_2 - f^2,\\
R_{16} &= V + c - fo,\\
R_{17} &= V - jc + k - 1 - hE,\\
R_{18} &= g + V^2 - y^2.
\end{aligned} \tag{4.2}$$

Finally, put

$$\begin{aligned}
R_{19} &= I - (2r+1)C_2, & R_{20} &= B_* - I^2,\\
R_{21} &= B_* - \Delta(F_2 - 1), & R_{22} &= y^2 - B_* g - 1,\\
R_{23} &= w - 4u^2, & R_{24} &= q - 2\rho^2,\\
R_{25} &= F\alpha - \mathrm{Pos}(A_1, A_2, A_3),\\
R_{26} &= FC - \mathrm{Pos}(B_1, B_2, B_3),\\
R_{27} &= \eta\zeta - \mathrm{Pos}(T_1, T_2, T_3).
\end{aligned} \tag{4.3}$$

The required single equation is

$$\boxed{Q_{42} = \sum_{i=1}^{27} R_i^2 = 0.} \tag{4.4}$$

Every residual has degree at most two in the supplied unknowns. The coordinate τ occurs only in R_8, with term τ^2, so the coefficient of τ^4 in Q_{42} is exactly one. This proves exact degree four on the authentic fixed-coefficient slice, independently of any generic degree-propagation estimate.

The literal JSON uses `v_scale`, `Baux` and `q_half_root` for v, B_*, ρ, and named triples for the three positive forms. The verifier expands all twenty-seven formulas above and checks exact equality with that JSON. In particular, no division, variable exponent or unspecified coefficient appears in the definition.

4.3 Preliminary guards and restored arithmetic

For every integer triple, $\mathrm{Pos}(x, y, t) > 0$, since $t(t+1) \geq 0$. Conversely every positive integer n has such a representation. Indeed $4n - 3 \equiv 1 \pmod 4$ is not of the forbidden form $4^a(8b+7)$, so the three-square theorem represents it by three squares. Exactly one square is odd. Writing that square as $(2t+1)^2$ and the others as $(2x)^2, (2y)^2$ gives the claimed representation.

At an integer zero of (4.4), every residual vanishes. The dyadic guard gives

$$q = \beta J + 1 = 2\rho^2.$$

Since $\beta \geq 3$, $J < 0$ would make $q < 0$, while $J = 0$ would require $1 = 2\rho^2$. Thus $J > 0$ and, for the bound source tuple, $q \geq 16$. The other two retained guards give $F\alpha > 0$ and $FC > 0$. Hence F, α, C have the same nonzero sign; their sum is $q > 0$, so

$$F > 0, \qquad \alpha > 0, \qquad C > 0. \tag{4.5}$$

Put $s = 2v + 1$, a nonzero odd integer, and set $Y = sq^3$ as a *proof abbreviation.* The first residuals give

$$N = q^3, \quad X = wq, \quad a = E + Y, \quad E = XY, \quad c = kY + \eta, \quad P = XY^2k + k,$$
$$H = 4a + 3, \quad \Delta = (a + 2)^2 - 1, \quad D = X + ac + \gamma H, \quad I = (2r + 1)c^2.$$

For example $R_5 = 0$ and $a = E + Y$ imply $E(X + 1) = X(E + Y)$, hence $E = XY$ with no division or nonzero assumption. Define the proof-only packing integer

$$R = \big[(q - 1)(q - F) + \alpha\big](q^2 - 1) + (M_C + qM_F)J. \tag{4.6}$$

The retained equations then assert

$$\tau^2 - XY^2(XY^2 + 1)k^2 = 1, \qquad D^2 - \Delta c^2 = 1, \tag{4.7}$$
$$I^2 = \Delta(f^2 - 1), \qquad I^2V^2 - (I^2 - 1)y^2 = 1, \tag{4.8}$$
$$k = R + 1 + hE, \qquad V = fo - c = jc - R. \tag{4.9}$$

The last two identities expose the quotient congruences directly; an unproved quotient-integrality assertion is unnecessary.

The source packing bounds require only (4.5) and the authentic mask bounds at this stage. With $G = q\alpha + (q - 1)C$, one has

$$q \le G \le q^2 - q, \qquad 0 < (M_C + qM_F)J < (q - 1)(1 + 2q).$$

Consequently

$$q(q^2 - 1) < R < q^4 - 2, \qquad R - 2 > 3q + 1. \tag{4.10}$$

No positivity assumption on z, η, ζ or Y has been used in obtaining these bounds.

4.4 Bootstrap with an unknown common sign

The new guard says $\eta\zeta > 0$. Let

$$\sigma = \operatorname{sign}(\eta) = \operatorname{sign}(\zeta), \qquad \kappa = |k| > 0, \qquad \theta = \frac{\eta}{k} \in (0, 1).$$

The fraction θ is used only in the proof. First $u \neq 0$. Otherwise $w = X = E = 0$, $a = Y$ and $D = ac + \gamma H$. Reducing the main Pell norm modulo H would give $0 \equiv 1 \pmod H$, since $\Delta \equiv a^2 \pmod H$. But $|Y| \ge q^3$ makes $|H| = |4Y + 3| > 1$. This is impossible. Therefore

$$w = 4u^2 \ge 4, \quad X \ge 4q > 0, \quad |Y| \ge q^3, \quad e := |E| \ge 4q^4 > 4R + 8. \tag{4.11}$$

Set $A = |a + 2|$, $P_0 = 2XY^2 + 1$, and $c_0 = |c|$. Then

$$A > e > R + 2, \qquad P_0 > A \ge 2, \qquad \Delta = A^2 - 1.$$

For negative Y, the first inequality follows from $A = (X + 1)|Y| - 2 > e$; for positive Y it is immediate. Moreover

$$\frac{c_0}{\kappa} = \begin{cases} |Y| + \theta, & Y > 0, \\ |Y| - \theta, & Y < 0, \end{cases}$$

so $c_0 > \kappa > 0$ in either case.

Let $\chi_A(n), \psi_A(n)$ be the integer Pell sequences defined by

$$\chi_A(n) + \psi_A(n)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^n.$$

Ordinary Pell classification in (4.7) gives positive ranks n, p with

$$\kappa = 2\psi_{P_0}(n), \quad |\tau| = \chi_{P_0}(n), \qquad c_0 = \psi_A(p), \quad |D| = \chi_A(p). \tag{4.12}$$

The first fundamental unit is the standard $2b + 1 + 2\sqrt{b(b+1)}$, where $b = XY^2 > 1$. Since $P_0 > A$ and $c_0 > \kappa$, strict Pell growth gives $p > n$. Also $P_0 \equiv 1 \pmod{e}$, so the recurrence gives

$$\sigma\, 2n \equiv R + 1 \pmod{e}. \tag{4.13}$$

If $\sigma = 1$, its least positive representative gives $2n \geq R+1$. If $\sigma = -1$, it gives $2n \geq e-R-1 > R+1$. Thus

$$n \geq (R+1)/2 \geq 24, \qquad p > n.$$

In particular the elementary Pell bounds imply

$$c_0 > A\Delta^2, \qquad c_0 > 2p, \qquad c_0 > 2R. \tag{4.14}$$

For the first, $p \geq 6$ already suffices, since $\psi_A(6) = 32A^5 - 32A^3 + 6A > A(A^2-1)^2$. The second follows by the recurrence for $A \geq 2, p \geq 3$; the third follows from $c_0 \geq \psi_A(2) = 2A > 2R$.

4.5 Recovering the rank without a sign assumption on c

We use the coefficient-independent ordinary strong-rank lemma retained with the source proof. In its needed form, if

$$c_0 = \psi_A(p) > A(A^2-1)^2, \qquad (ic_0^2)^2 = (A^2-1)(f^2-1), \quad i > 0,$$

then there is a positive integer m such that

$$|f| = \chi_A(m), \quad p \mid m, \quad c_0 \mid m, \quad m \geq c_0 > 2p, \quad ic_0^2 = (A^2-1)\psi_A(m). \tag{4.15}$$

Here its hypotheses hold with $i = |2r+1|$, using (4.8) and (4.14).

For clarity, the two substantive ingredients in that inherited lemma are strong divisibility and an elementary growth comparison. Write the quadratic field's fundamental integer-coordinate unit using a parameter F_0, and write $A = \chi_{F_0}(d)$, $S = \psi_{F_0}(d)$, and $\Delta = (F_0^2-1)S^2$. Classification yields $|f| = \chi_{F_0}(t)$ and $|I| = (F_0^2-1)S\psi_{F_0}(t)$. Strong divisibility, together with $c_0^2 \mid I$, gives $\psi_{F_0}(\gcd(t, dp)) \geq c_0/\Delta$. If that gcd were a proper divisor of dp, doubling its index would give

$$2\psi_{F_0}(\gcd(t,dp))^2 < \psi_{F_0}(2\gcd(t,dp)) \leq Sc_0 < Ac_0,$$

contrary to $c_0 > A\Delta^2$. Hence $dp \mid t$, giving $t = dm$ and $p \mid m$. Writing $m = pz$, the recurrence modulo c_0^2 gives $c_0 \mid \Delta z$. Finally $\gcd(c_0, \Delta) = \gcd(p, \Delta)$ follows from $\psi_A(p) \equiv pA^{p-1} \pmod{\Delta}$, and gives $c_0 \mid pz = m$. The archived ordinary-kernel proof supplies the complete signed and fundamental-unit details of this lemma.

The auxiliary norm in (4.8) excludes $V = 0$ and $y = 0$. Pell classification at the integer parameter $|I| > 1$ gives an odd rank $\ell = 2v_0 + 1$ such that

$$|I|\,|V| = \chi_{|I|}(\ell), \qquad |y| = \psi_{|I|}(\ell), \qquad |V| = Q_{v_0}(I^2).$$

Oddness follows from the even-index residue $\chi_{|I|}(2t) \equiv (-1)^t \pmod{|I|}$. The standard odd-subsequence polynomials satisfy

$$Q_{v_0}(1-A^2) = (-1)^{v_0}\psi_A(\ell), \qquad Q_{v_0}(0) = (-1)^{v_0}\ell.$$

They follow directly from $Q_0(x) = 1$, $Q_1(x) = 4x - 3$ and $Q_{t+1}(x) = (4x-2)Q_t(x) - Q_{t-1}(x)$.

The exposed identities in (4.9) imply

$$V \equiv -c \pmod{|f|}, \qquad V \equiv -R \pmod{c_0}.$$

As $I^2 \equiv 1 - A^2 \pmod{|f|}$, the first congruence gives $\psi_A(\ell) \equiv \pm c_0 \pmod{|f|}$, with all signs of c, V absorbed in the ambiguity. Squaring and applying the Pell doubling formula,

$$\chi_A(2\ell) \equiv \chi_A(2p) \pmod{\chi_A(m)}.$$

The signed step-down argument gives $\ell \equiv \pm p \pmod{m}$. Indeed reduce 2ℓ modulo $2m$, using $\chi_A(t + 2m) \equiv -\chi_A(t)$, then reflect indices above m. The resulting representative is $\pm\chi_A(s)$, $0 \le s \le m$. The case $s = m$ gives residue zero. A negative representative is impossible because $\chi_A(s) + \chi_A(2p) \le 2\chi_A(m-1) < \chi_A(m)$. Strict growth then forces $s = 2p$, proving the claimed index congruence. Its hypotheses $0 < 2p < m$ were proved in (4.15).

Since $c_0 \mid m$, this gives $\ell \equiv \pm p \pmod{c_0}$. On the other hand $c_0 \mid I$, so the second exposed congruence and $Q_{v_0}(0) = (-1)^{v_0}\ell$ give $\ell \equiv \pm R \pmod{c_0}$. Therefore $p \equiv \pm R \pmod{c_0}$. The strict bounds $0 < p, R < c_0/2$ in (4.14) force

$$p = R. \tag{4.16}$$

4.6 The missing signs and the reverse construction

If the common sign were negative, (4.13) would make e divide $2n + R + 1$. But $n < p = R$ and (4.11) give

$$0 < 2n + R + 1 < 3R + 1 < e,$$

a contradiction. Thus $\eta, \zeta > 0$. Represent them separately by Pos, and set the old square coordinate to $u_{\text{old}} = 2u$. Every residual of the authenticated predecessor now vanishes. Its remaining signed-scale and compiler arguments therefore produce a finite ZFC refutation. The missing individual guards have been restored before invoking that whole-system theorem.

There is a small transcription correction in the predecessor's written transport discussion. Its literal circuit, retained in R_{14}, has

$$(K_0 + w)C = F + z(q - 1),$$

whereas the prose writes $z - 1$. The needed conclusion is $z > 0$. After its Pell recovery, $w > q > F$ and $C \ge 1$ make the left side greater than F, hence $z(q-1) > 0$. This proves the required guard for the literal shifted coordinate. Neither the equation nor its coefficients are changed by correcting that sentence.

Conversely, a finite refutation supplies the predecessor's canonical accepting witnesses. Its complete coefficient adapter ensures

$$q = 2^t, \qquad t \text{ odd}, \qquad R \text{ odd}, \qquad w = 2^{R-t}, \qquad R > 3q.$$

Thus $R - t$ is even and at least four. Choose

$$u = 2^{(R-t-2)/2},$$

so $w = 4u^2$. The canonical η, ζ are positive; represent their product by one Pos triple and keep the other coordinates. All twenty-seven new residuals vanish. This is an existence construction using canonical witnesses, rather than an invertible substitution on every predecessor tuple.

4.7 What has been checked

The new construction retains twenty-five predecessor residuals exactly, strengthens the square scale, and replaces two positive guards by one. It changes $(45, 28, 4)$ to $(42, 27, 4)$ in the order unknowns, residuals, and single-polynomial degree. With the four ports left symbolic, the exact expanded sum of squares has 391 terms, compared with 407 for the predecessor. This is a symbolic description count, not a claim about coefficient bit complexity or an optimal arithmetic circuit.

The main builder and a separate implementation expand the literal residuals in sparse integer polynomial arithmetic. They check variable counts and occurrence, the unchanged rows, the two modifications, exact degree four, and the coefficient one of τ^4. Both work under ordinary and optimized Python. A separate transcription check verifies all twenty-seven factored formulas above against the emitted JSON. The parent polynomial and its proof are authenticated by their pinned Git blob hashes; the complete coefficient tuple and semantic source dependencies are retained with their own provenance.

These finite checks establish the displayed algebra and source binding. The universal solvability equivalence rests on the two-way mathematical argument and its expressly identified inherited arithmetic and compiler theorems. No consistency assumption, refutation search, global minimality assertion or external peer review is being substituted for that proof.

5 A sextic in thirty-five integers and an octic in thirty

The same fixed coefficient tuple also gives two equations with fewer unrestricted integer unknowns:

equation	unknowns	residuals	exact degree
S_{35}	35	20	6
O_{30}	30	15	8

Both satisfy the same ZFC-provable equivalence between absence of integer zeros and Con(ZFC). The collection's historical sextic and octic entries use forty-two and thirty-six unrestricted integers, respectively. These are improvements of seven and six unknowns relative to those entries, without a global minimality claim.

5.1 Move a square guard to the coordinate that will be eliminated

Let Q_{42}^X have the same residuals as Q_{42} except that

$$q - 2\rho^2 = 0 \qquad \text{is replaced by} \qquad X - 8\xi^2 = 0.$$

This replaces one unknown by one unknown. The retained equations include $w = 4u^2$ and $X = wq$.

Square-guard lemma. The integer zero sets of Q_{42} and Q_{42}^X are in bijection under $\xi = u\rho$, with all other coordinates fixed.

Proof. The equations $F\alpha > 0$ and $FC > 0$ imply that F, α, C have the same nonzero sign. Thus $q = F + \alpha + C$ satisfies $|q| \geq 3$, before either square guard has been used. If $u = 0$, then $w = X = 0$, the row $E(X + 1) = Xa$ gives $E = 0$, and

$$a = (2v + 1)q^3, \qquad |a| \geq 27.$$

With $H = 4a + 3$, the main norm has $D = ac + \gamma H$ and $\Delta = a^2 + 4a + 3 \equiv a^2 \pmod{H}$. Hence $D^2 - \Delta c^2 = 1$ implies $H \mid 1$, contrary to $|H| \geq 105$. Therefore $u \neq 0$.

At a zero of Q_{42}^X, the scale equations give $qu^2 = 2\xi^2$. Write $\xi/u = A/B$ in lowest terms with $B > 0$. Then $qB^2 = 2A^2$, so $B^2 \mid 2$ and $B = 1$. Consequently $\rho = \xi/u$ is an integer and $q = 2\rho^2$, restoring the deleted guard. Conversely, a zero of Q_{42} has $X = 4u^2 \cdot 2\rho^2 = 8(u\rho)^2$. The two substitutions are inverse. □

This elementary relocation matters for degree: $X = 8\xi^2$ is quadratic, while direct substitution of $w = 4u^2$ into $X = wq$ would give a cubic expression for X. The primary quartic remains the one in Section 4.

5.2 The thirty-five-integer sextic, in full

Keep the abbreviations $q = \beta J + 1$, $C = q - F - \alpha$, $k = \eta + \zeta$, and put $s = 2v + 1$. The twenty-six arithmetic unknowns are

$$J, F, \alpha, z, f, h, r, v, \tau_*, \eta, \zeta, y, \gamma, o, j,$$
$$E, c, a, D, L, V, I, B_*, \Delta, u, \xi.$$

Together with the nine guard unknowns from Section 4, there are thirty-five unknowns, all ranging over $\mathbb{Z}$. Here D, L, B_*, Δ remain independent coordinates. The twenty residuals are

$$\begin{aligned}
S_1 &= L - q^2 + 1, & S_2 &= 8\xi^2 - 4u^2 q,\\
S_3 &= a - E - sq(L+1), & S_4 &= E(8\xi^2+1) - 8\xi^2 a,\\
S_5 &= c - \eta - (a-E)k,\\
S_6 &= \tau_*^2 + (2\tau_* - k)E(c-\eta) - 1,\\
S_7 &= D - 8\xi^2 - ac - \gamma(4a+3),\\
S_8 &= \Delta - a^2 - 4a - 3, & S_9 &= D^2 - \Delta c^2 - 1.
\end{aligned} \tag{5.1}$$

The other residuals are

$$\begin{aligned}
S_{10} &= k - 1 - hE - \big[((q-1)(q-F) + \alpha)L + (M_C + qM_F)J\big],\\
S_{11} &= (K_0 + 4u^2)C - F - z(q-1),\\
S_{12} &= V + c - fo, & S_{13} &= V - jc + k - 1 - hE,\\
S_{14} &= I - (2r+1)c^2, & S_{15} &= B_* - I^2,\\
S_{16} &= B_* - \Delta(f^2 - 1), & S_{17} &= y^2 - B_*(y^2 - V^2) - 1,\\
S_{18} &= F\alpha - \mathrm{Pos}(A_1, A_2, A_3),\\
S_{19} &= FC - \mathrm{Pos}(B_1, B_2, B_3),\\
S_{20} &= \eta\zeta - \mathrm{Pos}(T_1, T_2, T_3).
\end{aligned} \tag{5.2}$$

The equation is

$$\boxed{S_{35} = \sum_{i=1}^{20} S_i^2 = 0.} \tag{5.3}$$

Every residual has degree at most three. Only S_{14} contains r, and the coefficient of r^2c^4 in S_{35} is exactly four. Thus its degree is exactly six after the authentic coefficients are substituted.

To derive this system, first change the root coordinate by

$$\tau_{\mathrm{old}} = \tau_* + P - k,$$

an invertible integral linear change. Then eliminate the seven registers

$$\begin{array}{c|l} P & k+E(c-\eta) \\ F_2 & f^2 \\ g & y^2-V^2 \\ X & 8\xi^2 \\ N & q(L+1) \\ C_2 & c^2 \\ w & 4u^2. \end{array}$$

Each register has a defining equation with coefficient 1 or -1. After substitution, the old root restores as $\tau_{\mathrm{old}} = \tau_* + E(c-\eta)$. The first norm becomes

$$[\tau_* + E(c-\eta)]^2 - [k+E(c-\eta)]E(c-\eta) - 1 = \tau_*^2 + (2\tau_* - k)E(c-\eta) - 1.$$

The cancellation of the fourth-degree term is exact. Every zero of the twenty-residual system uniquely restores a zero of Q_{42}^X, and every zero of Q_{42}^X yields one in the other direction.

5.3 The thirty-integer octic, in full

The twenty-one arithmetic unknowns are now

$$J, F, \alpha, z, f, h, r, v, \tau, \eta, y, \gamma, o, j, E, c, a, V, I, u, \xi.$$

Add the same nine guard unknowns. All thirty range over $\mathbb{Z}$. The following are polynomial abbreviations, not independent unknowns:

$$\begin{gathered} q = \beta J + 1, \quad C = q - F - \alpha, \quad s = 2v + 1, \quad H = 4a + 3, \\ k = 1 - V + jc + hE, \quad b = E(c-\eta), \quad \Delta = a^2 + 4a + 3, \\ R = \big[(q-1)(q-F) + \alpha\big](q^2-1) + (M_C + qM_F)J. \end{gathered}$$

The fifteen residuals are

$$\begin{aligned} O_1 &= 8\xi^2 - 4u^2 q, & O_2 &= a - E - sq^3, \\ O_3 &= E(8\xi^2+1) - 8\xi^2 a, & O_4 &= c - \eta - (a-E)k, \\ O_5 &= \tau^2 - (k+b)b - 1, & & \\ O_6 &= (8\xi^2 + ac + \gamma H)^2 - \Delta c^2 - 1, & & \\ O_7 &= -V + jc - R, & O_8 &= (K_0 + 4u^2)C - F - z(q-1), \\ O_9 &= V + c - fo, & O_{10} &= I - (2r+1)c^2, \\ O_{11} &= I^2 - \Delta(f^2-1), & O_{12} &= y^2 - I^2(y^2 - V^2) - 1, \\ O_{13} &= F\alpha - \mathrm{Pos}(A_1, A_2, A_3), & & \\ O_{14} &= FC - \mathrm{Pos}(B_1, B_2, B_3), & & \\ O_{15} &= \eta(k-\eta) - \mathrm{Pos}(T_1, T_2, T_3). & & \end{aligned} \tag{5.4}$$

The single equation is

$$\boxed{O_{30} = \sum_{i=1}^{15} O_i^2 = 0.} \tag{5.5}$$

Every residual has degree at most four. In O_{11} the term $-a^2 f^2$ has coefficient -1. No other residual contains both a and f, so the coefficient of $a^4 f^4$ in O_{30} is exactly one. The degree is therefore exactly eight on the fixed-coefficient slice.

Here the old root coordinate τ is retained. The twelve eliminated coordinates restore, after composing the substitutions, as follows:

N	q^3	C_2	c^2
F_2	f^2	g	$y^2 - V^2$
w	$4u^2$	X	$8\xi^2$
ζ	$k - \eta$	B_*	I^2
D	$8\xi^2 + ac + \gamma H$	Δ	$a^2 + 4a + 3$
P	$k + b$	L	$q^2 - 1$.

The actual monic elimination order is $N, C_2, F_2, g, w, X, \zeta, B_*, D, \Delta, P, L$. In particular, ζ is first removed using $\zeta = P - E(c - \eta) - \eta$, and P is later removed using the exposed congruence row. This yields the quadratic abbreviation for k, so the common-sign guard still has degree only three.

Each elimination is an exact integral bijection on zero sets. The square guard lemma then restores the original Q_{42} system. Applying its theorem proves the claimed consistency equivalences for both companions.

Reproducibility and scope. The companion generator checks nineteen monic eliminations, all fifty-four parent-row restoration identities, all thirty-five factored residual transcriptions, and both exact-degree coefficient certificates. It uses explicit exceptions rather than removable Python assertions. Normal and optimized runs agree. The literal and expanded JSON packets preserve the whole coefficient binding; no new coefficient transfer is assumed. These finite algebra checks supplement the ordinary proof. They do not constitute a proof-assistant certificate for the inherited unbounded Pell and compiler theorems. The JSON uses `rho` for ξ and `tau_tail` for τ_*.

6 A compact nine-positive companion

The supplied companion polynomial has nine strictly positive unknowns, exact total degree 1 834 516, and a live arithmetic circuit of 453 binary additions, subtractions and multiplications. At the inspected ProveIt revision, package59's corresponding point is $(9, 1\,910\,404, 473)$. Thus this component saves 75 888 degrees and 20 operations. It also binds the source to the complete compact44 coefficient tuple, whose four unsigned binary encodings occupy 295 388 bytes, compared with 533 099 for the preceding tuple. These are distinct measures: the constants remain large, and operation count is not bit complexity. This is a local Pareto improvement, not a claim of optimality or a major advance in universal variable counts.

The authoritative equation is the ordinary integer polynomial represented by the supplied arithmetic DAG:

`nine/generated/split_116_positive_global.json`.

Its fixed leaves are completely materialized and authenticated by `nine/fixed/BINDING.json`. Intermediate register names below are abbreviations, not quantified variables. In particular, all displayed exponents are fixed integers. The nine inputs, in order, are

$$b, g, h, s, w, C, i, j, n \in \mathbb{Z}_{>0}.$$

Using the identified inherited native compiler, its compact whole-tuple adapter, the calibration from K215 to finite refutations and the Jones ratio theorem, ZFC proves

$$(\exists b, g, h, s, w, C, i, j, n > 0\ \mathcal{F} = 0) \quad \Longleftrightarrow \quad \neg\,\mathrm{Con}(\mathrm{ZFC}).$$

The added reduction is ordinary mathematics formalizable in ZFC. The package supplies exact finite certificates, not a new end-to-end Lean/Rocq proof or an exported object-language ZFC derivation.

6.1 The source specialization and exact target certificate

Use the 31-positive-coordinate, 22-row source in `nine/data/parent_system60.json`. Its four symbolic coefficient ports are β, K_0, M_C, M_F. The complete compact44 tuple has bit lengths

$$(590069, 592884, 590069, 590070),$$

with aliases $K_0 = K$ and $M_F = M_{F,\mathrm{source}}$; the last port is already shifted by the native base minus one. The four payloads are supplied in `nine/coefficients/`. The supplied native builder reconstructs them exactly from the literal transition data. The finite verification includes all 66062 native basis identities and the explicit rank controller.

The old package59 source must be adjusted before this tuple is used. With all four ports symbolic, exact expansion gives

$$r_{15}^{\mathrm{new}} = r_{15}^{\mathrm{old}} - \beta J, \qquad r_i^{\mathrm{new}} = r_i^{\mathrm{old}} \quad (i \neq 15).$$

This is the justified transport shift $z_{\mathrm{old}} = z + 1$; its completeness uses the inherited source bound $z_{\mathrm{old}} \geq 2$ on zeros. It is not an arbitrary positive-coordinate substitution. Apply the invertible row operations

$$\rho_{14} = r_{14} + r_2, \qquad \rho_{15} = r_{15} + r_3, \qquad \rho_i = r_i \ (i \notin \{14, 15\}).$$

The resulting source has 89 records: 7 constant, 39 linear and 43 quadratic. In its original variable order choose weights

$$\begin{aligned}(w_i) = (&1, 7, 61, 36, 16, 85, 20, 67, 43, 11, 34, 13, 96, 80, 29, 50,\\ &2, 109, 110, 112, 116, 91, 53, 55, 27, 101, 92, 9, 107, 37, 60).\end{aligned}$$

Assign target exponents in the following paired order:

$$\begin{aligned}\text{rows} &: (2, 15, 8, 17, 13, 18, 12, 20, 21, 4, 22, 9, 11, 19, 1, 6, 10, 16, 3, 14, 7, 5),\\ \text{targets} &: (28, 124, 325, 423, 598, 658, 775, 904, 941, 955, 1027, 1236,\\ &\quad 1245, 1313, 1323, 1530, 1571, 1584, 1637, 1787, 2014, 2036).\end{aligned}$$

Write q_j for the target of row j, and set $U_y(T) = \sum_i y_i T^{w_i}$. For a coefficient $a_{j,\alpha}$ of y^α in ρ_j, let $d = |\alpha|$ and $e(\alpha) = \sum_i \alpha_i w_i$. Define

$$c_{j,\alpha} = \begin{cases} 2a_{j,\alpha}, & d = 0, 1,\\ a_{j,\alpha} \prod_i \alpha_i!, & d = 2, \end{cases} \qquad D_d(T) = \sum_{j,\,|\alpha|=d} c_{j,\alpha} T^{q_j - e(\alpha)}.$$

The supplied literal sparse kernels and target certificate prove

$$[T^{q_j}](D_2(T)U_y(T)^2 + D_1(T)U_y(T) + D_0(T)) = 2\rho_j(y) \quad (1 \leq j \leq 22).$$

Indeed, among all 22544 same-degree source/expansion interactions, only the 89 intended contributions hit selected targets. Squares have multiplicity one in U_y^2 and cross terms multiplicity two, so the stated weighting recovers twice the source coefficient. This is a formal coefficient identity, with arbitrary witnesses and arbitrary coefficient ports. The resulting degrees are

$$(\deg D_0, \deg D_1, \deg D_2) = (1571, 1934, 1836), \quad W = 116, \quad H = \max_d(\deg D_d + dW) = 2068.$$

The leading coefficient of D_2 is exactly $+1$. There are no negative kernel exponents. The maximal target is 2036, leaving the required headroom below the convolution height.

6.2 Source words and the exact binary threshold

Replacing native-port powers by their strict bit bounds and summing absolute values gives the sharp common coefficient majorant $M = 2^{1182954}$: every actual weighted coefficient has magnitude strictly less than M. Set

$$\kappa = 1182966, \quad a = 2^{\kappa} = 4096M, \quad B = ab^2, \quad \ell = \sum_i B^{w_i}, \quad U_s = \ell + g.$$

Define the following integer polynomials:

$$\begin{aligned} G_s &= (D_2(B)U_s + D_1(B))U_s + D_0(B), & T_* &= \sum_j B^{q_j}, \\ N_0 &= B^{117}, & Z &= B^{2069}, \\ \mathrm{Mask} &= N_0 - 1 - (b-1)\ell, & J_* &= 2G_s + T_*, \\ S_* &= g + N_0 J_*, & T_{**} &= \mathrm{Mask} + N_0(B-2)T_*. \end{aligned}$$

For every $b, g \geq 1$, we have $U_s \geq B^{116}$ and $D_2(B) > B^{1836}/2$. The two potentially negative contributions, relative to $B^{1836}U_s^2$, are bounded by

$$\frac{|D_1 U_s|}{B^{1836}U_s^2} < \frac{2M}{B^{18}} < \frac{1}{8}, \qquad \frac{|D_0|}{B^{1836}U_s^2} < \frac{2M}{B^{497}} < \frac{1}{8}.$$

Thus $G_s > 0$ unconditionally; also $0 \leq \mathrm{Mask} < N_0$, so the source words are nonnegative before any finalizer consequence is used. On the full soundness range $g < 4bB^{116}$, one has $U_s < 6bB^{116}$ and

$$\frac{2G_s}{Z} < \frac{86Mb^2}{B-1} < \frac{1}{32}, \qquad T_* < \frac{Z}{64}, \quad (B-2)T_* < \frac{Z}{64}, \quad g < N_0.$$

Consequently $S_* + T_{**} < B^{2186}$. For binary b, the low mask forces $g = \sum_i z_i B^{w_i}$ with $0 \leq z_i < b$, so U_s encodes the positive witnesses $y_i = z_i + 1$. Every convolution coefficient is bounded in magnitude by $993Mb^2 < B/4$. The inherited signed-borrow padding argument, now with the certified target identities and these numerical bounds, therefore gives

$$S_* \,\&\, T_{**} = 0 \quad \Longleftrightarrow \quad g \text{ encodes a positive common source zero.}$$

A positive source zero supplies such a word for every sufficiently large binary b, with $0 < g < bB^{116}$. The strict positivity follows because the all-one source assignment has $r_5 = 1$.

Now put

$$L = bB^{2186}, \quad R = L(S_* + T_{**} + 1) - S_* - 1, \quad N = 2^{8872234}B^{1155}.$$

Unconditionally $R, N \geq 8$ and $b \leq R, N$. If $b = 2^k$, write $v = 1182966 + 2k$ and $u = k + 2186v$, so $B = 2^v$ and $L = 2^u$. The bounded source words give the two-block expansion

$$R = (S_* + T_{**})L + (L - 1 - S_*).$$

With pop denoting binary population count and $\tau = \mathrm{pop}(S_*) + \mathrm{pop}(T_{**}) - \mathrm{pop}(S_* + T_{**})$, this implies

$$\mathrm{pop}(R) = u + \mathrm{pop}(T_{**}) - \tau.$$

The low mask contributes $117v - 31k$ bits and the 22 high digits contribute $22v - 22$ bits. Hence the exact calibration is

$$u + \mathrm{pop}(T_{**}) = 2325v - 30k - 22 = 2(8872234 + 1155v) = 2\log_2 N.$$

Since $\tau \geq 0$, with equality exactly for binary-disjoint words, Legendre's identity $v_2\binom{2R}{R} = \mathrm{pop}(R)$ proves

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_* \,\&\, T_{**} = 0.$$

The case $k = 0$ is included and cannot produce a final zero, since then $\mathrm{Mask} = N_0 - 1$ and $0 < g < N_0$.

6.3 The explicit final equation

The remaining formulas are the inherited free-coordinate finalizer, applied with the parameters just proved. Write $j_0 = j - 1 \geq 0$ and define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M_0 &= RY,\\
P &= 2M_0^2 U, & K &= R + 1 + h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R+1, & W_1 &= bw, & J &= 4A - 5,\\
Q &= 3W_1 C - 2(W_1^2 - 1), & & & D &= (A^2-1)C^2 + 1,\\
k_0 &= iJC, & E &= k_0 DC, & F_1 &= (A^2-1)E^2 + 1,\\
G &= A + F_1(CD + 1 - A), & & & H_1 &= C + F_1(B_1 + j_0 C),\\
I &= (G^2-1)H_1^2 + 1, & a_0 &= (P^2-1)K^2 + 1, & a_1 &= DF_1 I.
\end{aligned}$$

Finally set

$$\begin{aligned}
t &= bB^{116}, & m &= K^2(4t-g) - 16t(C-KY)^2,\\
V_0 &= PK, & V_1 &= k_0(8D^2 - 4D - 1)GH_1,\\
\Omega &= Jm, & E_0 &= J(n + W_1^2) + Q,\\
\Theta &= E_0 - \Omega(V_0 + V_1).
\end{aligned}$$

The equation is

$$\boxed{\mathcal{F} = [\Theta^2 - \Omega^2(a_0 + 64a_1)]^2 - 256\Omega^4 a_0 a_1 = 0.} \tag{6.1}$$

The inherited proof applies because $R, N \geq 8$, $b \leq R, N$, the source words are unconditionally nonnegative, and the source completeness bound $g < bB^{116}$ has the required factor-32 margin. The important positivity step occurs before square-root integrality:

$$V_0^2 - a_0 = K^2 - 1 > 0, \quad V_1^2 - 64a_1 = [(8D+1)k_0^2 - 64D](GH_1)^2 + 64DF_1(H_1^2 - 1) > 0.$$

Also $E_0 > 0$. A norm zero with $m = 0$ would give $E_0^4 > 0$; otherwise its real factorization forces $m > 0$. This yields $|C - KY| < K/2$ and the needed carrier bound. The rational-root separation then makes both square roots integral, gives $J \mid Q$, and invokes the Jones ratio theorem to recover binary b, the source range and binomial divisibility. Conversely the source and ratio witnesses yield an integer $n > 0$ by the explicit bound $W_1^2 + Q/J < 2D < V_1$. Thus no unproved cofinality claim about j or n enters the construction. The complete inherited argument and the new applicability checks are retained in `nine/PROOF.md` and its pinned upstream dependencies.

6.4 Exact degree, cost, and verification

The source heads give

$$\deg S_* = 4370, \quad \deg T_{**} = 4308, \quad r = \deg R = 8743, \quad q = \deg N = 2310.$$

Let $\Delta_1 = V_1^2 - 64a_1$. To resolve the apparent leading cancellation in the norm, use the exact identity

$$\Theta^2 - \Omega^2(a_0 + 64a_1) = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2\big[(K^2 - 1) + \Delta_1 + 2V_0V_1\big].$$

The separate degree certificate has a positive diagonal leading term and no unresolved positive-degree cancellation. The diagonal reaches the multivariate upper bound, proving the exact degree

$$104r + 400q + 8W + 316 = 1\,834\,516.$$

The initial Horner circuit has 467 operations. Its 102 pure powers of B are replaced by a shared 88-gate addition chain. Every chain row $e = a + b$ certifies $B^e = B^a B^b$, and exact structural normalization checks the complete rewritten circuit, including the definition of B. The final tally is 162 additions, 22 subtractions and 269 multiplications, totaling 453. Scalar multiplication and every power multiplication are included; constant leaves and reuse are free.

Replacing every positive input by a nonnegative input plus one gives the supplied nine-nonnegative companion, with the same degree and 460 operations. No nine-variable unrestricted-integer claim is made. The 12 final nonliteral leaves occupy 2 217 161 raw magnitude bytes or 47 364 deterministic gzip bytes; the signs and hashes are explicit.

The command `python3 nine/reproduce.py` reconstructs the entire component offline using the Python standard library. It regenerates the source and both domains, verifies all target identities, reconstructs the native tuple, materializes the derived leaves, and checks the power-chain and exact-degree certificates. A second finite checker imports no project builder or verifier. All authored checks use exceptions and remain enabled under `python3 -O`. Modular and small binary-block fixtures are separately labeled as finite transcription checks; they do not establish the unbounded consistency semantics.

7 Verification, source correction, and reproducibility

The archive contains the final polynomial definitions, their fixed coefficient bindings, the mathematical proofs, and offline verification programs. Its new checks use the Python standard library. Historical scripts that require other packages are preserved only as source evidence when relevant; the new replay does not need to execute them. The root command is

```
python3 verify_all.py
```

Run it from the extracted package directory. It authenticates the package, executes the component checks, reconstructs the bound native coefficient tuples, and replays the finite polynomial and circuit certificates. The receipt is `verification/REPLAY.json`. Both ordinary and optimized Python execution were used during preparation; acceptance does not rely on removable `assert` statements.

For Q_{42}, two separate sparse polynomial implementations check the literal residuals, the $45 \to 42$ modification, all variable occurrences, the exact degree, and the coefficient one of τ^4. A transcription check compares the displayed factored equations with the JSON. The coefficient replay regenerates all four source-34 numerals and checks 69 230 native basis conditions and 8 801 coefficient digits. The companions have explicit restoration maps after the proved scale-guard exchange: deleting coordinates through monic equations preserves the integer solution set by substitution, with no rational divisions.

For P_{13}, the certificate separates a multivariate degree upper bound from a nonzero homogeneous leading value, so a sampled univariate degree is not mistaken for an upper bound. The norm identity, unit-product residues, complete operation ledger, and fixed coefficient binding are checked. The nine-variable replay verifies all 22 544 sparse target interactions, the coherent native source change, exact degree data, every shared power identity, and the complete circuit rewrite. It checks all 66 062 native basis identities of the compact tuple. These are finite certificates for exact algebra and data. They do not replace the unbounded two-way compiler and Pell arguments.

One source correction is necessary. The predecessor quartic's prose writes the transport coordinate as $z - 1$, whereas its authenticated literal equation is

$$(K_0 + w)(q - F - \alpha) = F + z(q - 1).$$

The correct required restored guard is $z > 0$. The old positive product coordinate would be $z + 1$. After the source Pell recovery, $w > q > F$ and $q - F - \alpha \geq 1$ prove this strict positivity directly. The new quartic proof states the correction; the original source file is retained unchanged for comparison. Neither the literal polynomial nor its coefficients are silently repaired to match the prose.

The reviews included in the archive are internal independent-agent mathematical or implementation reviews in this research session. They are identified as such. They are not external human peer review, and finite diagnostics are labeled separately from universal proofs.

7.1 File map

Path	Purpose
`report.tex`, `report.pdf`	This article and its editable source.
`positive13/`	The degree-114 polynomial, complete norm proof, fixed tuple binding, degree and circuit certificates.
`quartic42/`	All 27 quadratic residuals of the quartic, common-sign proof, exact expansion and independent algebra check.
`quartic42/companions/`	The sextic and octic, coordinate restoration data and exact elimination certificates.
`quartic42/dependencies/`	Authenticated ordinary compiler and Pell dependencies, source-34 coefficient payloads and their replay.
`nine/`	The optimized nine-variable circuits, sparse code and kernel certificates, compact native tuple and final fixed leaves.
`RESULTS.json`	Machine-readable domain, variable, degree and operation ledger for the new forms.
`SOURCE_SNAPSHOT.json`	Comparison revision and source locations.
`MANIFEST.json`	SHA256 hashes of delivered files.
`verify_all.py`	One-command offline finite verification.
`RESEARCH_NOTES.md`	Unproved directions and rejected shortcuts, kept separate from the theorems.

Compile the article with

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The PDF was rendered and inspected after compilation. The source contains the complete formulas rather than relying on remote images or an online notebook. The archive includes the upstream licenses and notices; no external repository or publication is modified by this contribution.

8 Where a larger advance could come from

The new reductions suggest specific further work, but they do not yet resolve the request for a major change of scale. The two clearest new mechanisms are reusable: recover a missing sign from an already available rank modulus, and center a root before eliminating it by a norm. Both exploit constraints already paid for elsewhere in the equation.

Recover more signs without duplicating positive forms. The quartic argument saves three witnesses because its common-sign ambiguity is a single bit that the index congruence can decide. A further reduction would need a comparable argument for other guards. Removing them on the strength of the old whole-system theorem would be circular: that theorem assumes those very guards. In particular, simply allowing both scale coordinates to be negative changes the main binomial recovery and has not been justified here.

Use a smaller direct binary-disjointness interface. The nine-variable companion spends most of its degree converting *binary disjointness* of two packed nonnegative integers S, T into a divisibility threshold on a central binomial coefficient. The direct condition

$$\binom{S+T}{S} \equiv 1 \pmod 2$$

expresses absence of binary carries. A sufficiently economical Diophantine interface for this condition could avoid the large two-block index and threshold used here. The carry interpretation by itself is not such an interface; a new explicit polynomial construction with a complete converse is still needed.

Combine root elimination with divisibility, not only radicals. Norm equations are attractive because they remove witnesses, but a rational square root need not satisfy a required quotient congruence. The thirteen-variable norm works because the first recovered root is monic over the integers and the other root is then an integer difference. Eliminating additional roots while losing one of the divisibility guards can admit extraneous solutions. For example,

$$63\left((4/3)^2 - 1\right) = 7^2$$

shows that the strong Pell-looking equation alone does not force a rational f to be integral. This is a counterexample to that proposed local shortcut, not to the full guarded source system. No smaller equation obtained by that unsupported cancellation is claimed.

Change the proof-checking source itself. A different primitive-recursive refutation checker, direct arithmetic verification of a proof certificate, or a different finite tiling language could reduce coefficient and compiler complexity. An appeal to MRDP alone does not supply the necessary literal coefficients, degree bounds, or two-way compiler proof. This package does not replace K215 by an unimplemented alternative.

The contributions ready for repository review are the explicit equations and reductions proved above. Further priority, global optimality, and proof-assistant certification require separate work.

References

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[6] ProveIt contributors, *Degree separated*, package 59, including the ordinary merger, thirteen-positive adjunct, and nine-free adjunct, pinned comparison commit `beb0192cb8dc36f78c369c080f2adfbc1887a486`. https://github.com/VladimirReshetnikov/ProveIt/tree/beb0192cb8dc36f78c369c080f2adfbc1887a486/Computability/DiophantineConsistency/con-zfc-diophantine-forms/59-degree-separated.

[7] ProveIt contributors, inherited coefficient, ordinary Pell, signed-scale, native compiler and K215 calibration sources. Exact historical revisions, repository paths, byte hashes and local reading map are supplied in `quartic42/dependencies/SOURCE_PINS.json`, `quartic42/dependencies/README.md`, and the component `SOURCE_PINS.json` files. Original licenses and notices accompany the authenticated copies.